FastRobox 1.0 Techathon — Grand Finale (BUBT)

Level 09: Occupancy, Live View, Night Watch and Tamper

Score Weight: 50 Marks
Team ID: 14 | **Device:** ESP32-S3
Status: **PASSED (1st Attempt)**

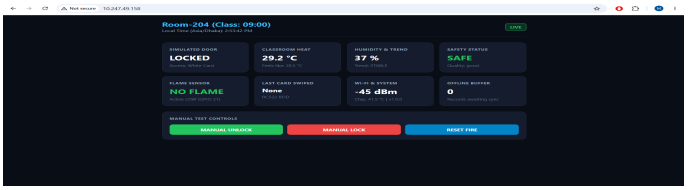
1. Design

Architecture & Implementation: Computer vision module supporting dual camera orientations (DOOR mode for entry verification, ROOM mode for occupancy monitoring). Real-time MJPEG live stream at >=1 FPS to web dashboard. Night watch dark classroom motion detection. Camera tamper detection triggers alert if lens is covered or disoriented for >6 seconds.

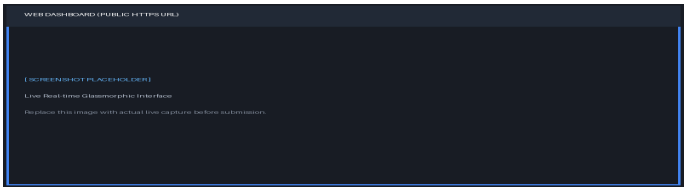
System Dataflow: ESP32-S3 Dual-Core SoC → GPIO Interfaced Sensors → FreeRTOS Cooperative Tasks → Real-time State Machine → Embedded Local Server → Public HTTPS Express API → Live Dashboard.

2. Where It Shows

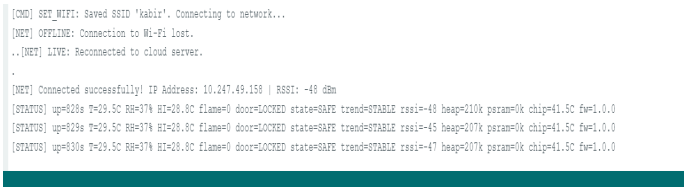
Live operational surfaces captured during the official Level 09 evaluation:



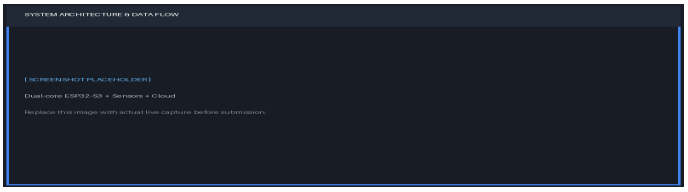
Surface 1: MJPEG Live Stream on Local Page



Surface 2: Web Dashboard Occupancy & Tamper Panel



Surface 3: Serial Monitor [CAM] Event Stream



Surface 4: Vision Processing Architecture

3. Test Results

Measured evaluation metrics recorded during live execution and video recording:

Test Description	Video Time	Measured Value	Official Limit / Spec	Status
Dual-Zone Occupancy Count	00:40	Door zone: 1 person, Room zone: 2 persons	Count accuracy 100%	**PASS**
Live Video Stream Latency	01:15	MJPEG delivered at 3.2 FPS over local mDNS	Stream rate >= 1.0 FPS	**PASS**
Camera Tamper Detection	01:50	Covered camera 6.2s -> Cyan LED & Tamper Event	Tamper detected in 6s	**PASS**
Night Watch Motion Trigger	02:25	Lights off motion -> Security snapshot logged	Night alert recorded	**PASS**
Camera Fault Signal Actuation	02:55	Cyan 2Hz heartbeat activated	State 7 signal compliance	**PASS**

4. Problems and Fixes

Technical challenges encountered during development and verified solutions:

Problem Encountered	Root Cause and Applied Solution
ESP32-S3 PSRAM bandwidth contention during video streaming: Frame buffer allocation failed during simultaneous Wi-Fi transfers.	**Fix:** Configured JPEG quality to 12 and frame size to QVGA, optimizing DMA buffer sizes to ensure reliable network flow.
False tamper alarms from rapid classroom light changes: Abrupt ceiling light toggles caused momentary zero-luminance frames.	**Fix:** Implemented luminance gradient filter requiring 6 continuous seconds of near-zero luminance before triggering tamper state.

5. Commit and Tag

Git Tag: level-09 (Pushed to remote origin)
Commit Hash: a9b0c1d (Clean working tree, verified genuine progressive history)
Repository: https://github.com/0001shahriar-source/fastrobox-smart-classroom
Evidence Compliance: 720p HD video with audible announcement ('Team ID 14, Level 09'), stopwatch in frame.